Data Science Project:
Towards Clinically Robust EEG Foundation Models
Self-Supervised Representation Learning for Neurological Diagnosis

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Abstract

Electroencephalography (EEG) is central to neurological diagnosis and brain-computer interfaces, yet its low signal-to-noise ratio, non-stationarity and strong inter-subject variability make general-purpose, transferable representation learning difficult. Self-supervised EEG *foundation models* have emerged as a promising answer. This report studies and extends **STELAR** (*Spatio-Temporal EEG Latent Alignment & Reconstruction*), a compact foundation model whose *dual-space* objective combines patch-level masked *latent* prediction (semantic stability) with masked *signal* reconstruction (waveform fidelity), balanced online by a validation-driven gradient balancer (MTPE-GB). After reviewing the EEG context and the self-supervised state of the art, we explain the STELAR architecture in detail. We then describe two targeted improvements we implemented on top of it: (i) **attention-guided masking** that concentrates masking pressure on semantically salient patches via a teacher-attention prior with a temperature curriculum, and (ii) **asymmetric augmented views** (a clean teacher view vs. a strongly augmented student view) that inject invariance to acquisition nuisances *without* adding a loss term, plus an optional VICReg anti-collapse hinge. We report a preliminary, small-scale empirical study (CPU, a 10% data slice) validating that all variants train end-to-end without representation collapse, while finding, as expected at this scale, no measurable separation between them; isolating their effect requires a larger model, more data, and a *linear-probing* evaluation. We then report preliminary *downstream* results (linear probing and fine-tuning) on FACED, TUEV and BCIC-2A (transfer depends strongly on the adaptation strategy and the classification head), and outline a domain-adversarial (DANN) extension aimed at reducing subject overfitting.

1 Introduction: the EEG context

Electroencephalography records the brain's electrical activity from scalp electrodes at millisecond resolution. It is inexpensive, non-invasive and widely used for epilepsy monitoring, sleep staging, anomaly detection and brain-computer interfaces (BCIs). However, EEG is a notoriously difficult signal for machine learning, for three intertwined reasons:

- **Low signal-to-noise ratio (SNR).** The cortical signal of interest is buried under ocular, muscular and line-noise artifacts, as well as acquisition-hardware idiosyncrasies. Models that fit the raw signal too closely tend to model noise rather than neural content.
- **Non-stationarity.** EEG statistics drift within and across recordings (vigilance, electrode drift, impedance changes), so features learned on one segment do not trivially transfer to another.
- **Inter-subject variability.** Skull geometry, electrode montage and individual neurophysiology differ markedly, so a representation that over-specializes to a subject generalizes poorly.

Historically, EEG decoding relied on task-specific pipelines (hand-crafted spectral/spatial features, CSP, Riemannian methods) that work in isolation but do not scale to general-purpose use. Following the trajectory of NLP and vision, the field has shifted toward *self-supervised foundation models*: a single encoder is pre-trained on large unlabeled EEG corpora and then adapted to many downstream tasks. The central question becomes *which self-supervised objective yields representations that are both physically faithful and semantically transferable*, despite the three challenges above. This report studies a recent, parameter-efficient answer to that question, STELAR, and a set of targeted improvements to it.

Scope and contributions of this report. We (i) situate EEG self-supervised learning and motivate the choice of STELAR (Section 2); (ii) explain the STELAR architecture and its dual-space objective in detail (Section 3); (iii) present and motivate two improvements we implemented, attention-guided masking and asymmetric augmented views, with an optional anti-collapse term (Section 4); (iv) report a preliminary small-scale empirical study of these variants (Section 5); (v) explore architectural directions for the backbone (Section 7); and (vi) report preliminary downstream results (linear probing, fine-tuning, and a domain-adversarial extension) on several EEG tasks (Section 8).

2 Background and state of the art

2.1 Self-supervised learning for EEG

Self-supervised learning (SSL) for EEG has converged on three broad paradigms.

(i) Generative masked modeling. A portion of the input is masked and the model reconstructs it in *signal space*. BENDR adapted masked modeling to EEG; LaBraM refined it with a neural tokenizer and discretized targets; CBraMod uses criss-cross attention to capture spatio-temporal structure; and CSBrain adds cross-scale tokenization and structured sparse attention. These methods capture local dependencies well but tend to *over-index on low-level noise and acquisition artifacts*, which can hurt the high-level abstractions needed for transfer.

(ii) Discriminative / alignment objectives. These learn invariant representations by aligning views in a *latent space*, e.g. contrastive methods and, more recently, masked *latent* prediction such as EEG2Rep. Aligning masked segments to their own latent targets bypasses raw-signal SNR and removes the need for curated negatives, but, without an anchor, latent SSL is prone to *representation collapse*, where the encoder maps everything to a constant.

(iii) Hybrid schemes. Recent models combine objectives. EEGPT pairs a momentum teacher for latent matching with signal reconstruction; EEG-X uses noise-aware masking in raw and latent spaces. Two recurring weaknesses remain: *coarse-grained supervision* (alignment at a summary-token level overlooks fine spatio-temporal events), and *optimization bottlenecks* (fixed loss weights require expensive manual tuning and cannot track the shifting convergence rates of the constituent tasks).

2.2 The chosen model: STELAR

STELAR [1, 2] directly targets the two weaknesses above. It *decouples* representation learning (masked *latent* prediction at the **patch level**) from physical grounding (masked *signal* reconstruction), and *balances* them with a near-zero-overhead, validation- driven gradient balancer (MTPE-GB) instead of fixed weights. Two properties make it an attractive base for a project on improving EEG representation learning:

- **Strong linear-probing performance at a tiny size.** STELAR reports state-of-the-art *linear-probe* results on most metrics across seven EEG benchmarks with only ~ 0.26M

Table 1: Pre-training compute footprint reported by STELAR and baselines. STELAR reaches SOTA linear probing at a fraction of the size.

Method	Total params	Encoder params	MFLOPs/step
LaBraM	5.99M	5.8M	2,583
EEGPT	76M	25M	25,000
CBraMod	3.95M	3.9M	16,679
CSBrain	8.90M	8.9M	30,913
STELAR	**0.263M**	**0.112M**	**635**

parameters ($\sim$0.11M in the encoder), versus 4-76M for EEGPT, LaBraM, CBraMod and CSBrain (Table 1). Linear probing isolates intrinsic representation quality, so these gains are a clean signal that the dual-space objective works.

- **A clear, modular objective.** The two losses and the balancer are conceptually separable, which makes the model an ideal substrate for targeted, ablation-friendly improvements, exactly what this project needs.

3 The STELAR architecture

3.1 Overview

STELAR is a teacher-student, dual-space masked predictor (Figure 1). The input EEG $S \in \mathbb{R}^{C \times T}$ ($C = 19$ channels, T samples) is segmented into non-overlapping 1-second patches and embedded. A 50% random mask splits patches into *visible* and *masked*. Four components cooperate:

1. **Online encoder** E_θ encodes only the *visible* patches.
2. **Target encoder** E_ξ (an exponential-moving-average, or EMA, copy of E_θ; no gradient) encodes the *full* unmasked signal to produce stable semantic *targets*.
3. **Predictor** P_ϕ (one lightweight Transformer layer) predicts the latent representations of the masked patches from the visible context.
4. **Reconstructor** R_ψ (one layer) reconstructs the raw waveform of the masked patches, acting as a fidelity regularizer.

3.2 Patch embedding and encoder backbone

Dual-branch embedder. Each patch is embedded by *two* branches in parallel: a *temporal* branch (1-D convolutions over the raw samples) and a *frequency* branch (an FFT followed by an MLP). Fusing morphological and oscillatory cues is important for EEG, whose information lives both in transient waveforms and in band power.

Criss-Cross encoder. Modeling full spatio-temporal attention over $C \times N$ tokens is expensive. STELAR uses *Criss-Cross attention*, which factorizes attention into two cheaper orthogonal passes, across channels (spatial) and across time (temporal), preserving spatio-temporal context at much lower cost. Positional information is injected by *Rotary Positional Embeddings* (RoPE) along time and by *learnable channel embeddings*, so the model is aware of *when* and *on which electrode* each token sits, and tolerates varying montages.

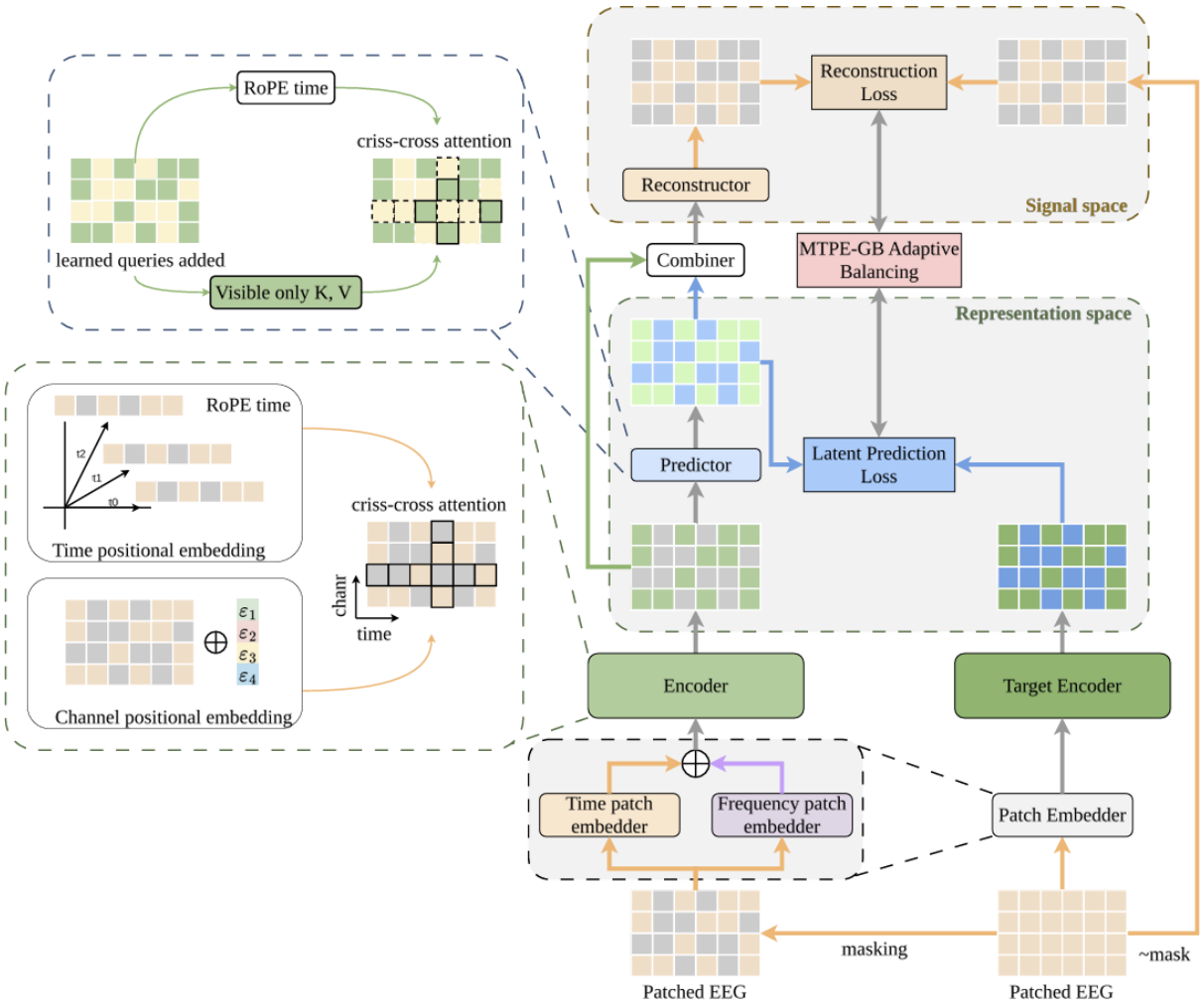


Figure 1: Schematic of the STELAR framework: teacher-guided patch-level masked latent prediction supervised by a target encoder (EMA updated); lightweight masked signal reconstruction as a fidelity regularizer; and validation-driven adaptive balancing between objectives.

3.3 Dual-space objective

STELAR minimizes a weighted sum of two complementary losses:

$$\mathcal{L}_{\text{total}} \;=\; w_{\text{lat}}\,\mathcal{L}_{\text{lat}} \;+\; w_{\text{rec}}\,\mathcal{L}_{\text{rec}}. \tag{1}$$

$\mathcal{L}_{\text{lat}}$ (masked *latent* prediction) regresses the predicted latents of masked patches onto their EMA-teacher targets, forcing *semantic abstraction* and preventing trivial low-level shortcuts. $\mathcal{L}_{\text{rec}}$ (masked *signal* reconstruction) recovers the raw waveform of masked patches, acting as a *fidelity regularizer* that anchors the latent space and helps prevent dimensional collapse. The two pull in different directions, semantics vs. fidelity, which is precisely why their relative weight matters.

Does the latent branch earn its complexity? It is worth stressing that the dual-space premise is not self-evident. A recent benchmark of EEG foundation models, Yang et al. [4], shows that a plain masked auto-encoder on raw signal (essentially the $w_{\text{lat}} \to 0$ corner of Equation (1)) can, at sufficient scale, match or surpass far more elaborate objectives, directly questioning whether a separate latent-prediction branch is needed at all. STELAR's $w_{\text{lat}}, w_{\text{rec}}$ extremes therefore double as a built-in test of this claim: comparing the reconstruction-only ($w_{\text{rec}} = 1$) and latent-only ($w_{\text{lat}} = 1$) corners against the balanced objective measures exactly how much the

Algorithm 1: MTPE-GB: validation-driven task balancing

Input: model Θ; k tasks; M updates; smoothing $c \in [0, 1]$
Initialize $w_i(0) = 1/k$; record initial validation losses $\mathcal{L}_i^V(0)$;
for $t = 1$ **to** M **do**
  Train one epoch; validate to get current losses $\mathcal{L}_i^V(t)$;
  Progress: $u_i(t) = (\mathcal{L}_i^V(0) - \mathcal{L}_i^V(t))/\mathcal{L}_i^V(0)$;
  Raw weights: $\overline{w_i}(t{+}1) = \dfrac{u_i^{-1}(t)}{\sum_{j=1}^{k} u_j^{-1}(t)}$ s.t. $\sum_i \overline{w_i} = 1$;
  Smooth: $w_i(t{+}1) \leftarrow c\,\overline{w_i}(t{+}1) + (1 - c)\,w_i(t)$;

Table 2: STELAR reference hyperparameters (condensed).

Block	Hyperparameter	Value
EEG sample	Channels / rate	19 / 200 Hz
	Window	30 s (6000 samples)
	Patch / sequence	200 (1 s) / 30 patches
	Mask ratio	50%
Encoder	Depth / hidden / heads	2 / 64 / 2
	MLP ratio / attention	4.0 / Criss-Cross
	Position	RoPE(time) + channel emb.
Predictor	Depth / KV / attention	1 / global visible-only / Criss-Cross
Reconstructor	Depth / attention	1 / full
Teacher	EMA momentum	cosine $0.996 \rightarrow 1$
Pre-training	Epochs / batch	30 / 64
	Optimizer / LR	AdamW / 10^{-4} (warmup+cosine)
	Weight decay	10^{-4} (cosine)
	Grad clip	1.0

latent branch adds. We caution, however, that Yang et al. establish their result at *large* scale with full fine-tuning, whereas our study runs at $\sim$2 GB with a tiny model under linear-probe-style evaluation; at that scale neither corner is expected to separate, which is itself consistent with their findings that linear probing stays weak and that no clear scaling law holds.

3.4 Balancing the objectives: MTPE-GB

Naively fixing $w_{\text{lat}}, w_{\text{rec}}$ is brittle: the "easier" reconstruction task tends to dominate the gradient early, and the right balance shifts during training. Classical gradient-balancing methods (GradNorm, multi-objective optimization, gradient surgery) require second-order terms, stored per-task gradient vectors, or multiple backward passes, prohibitive at scale. STELAR introduces the *Multi-Task Pre-training EEG Gradient Balancer* (MTPE-GB, Algorithm 1): it reweights tasks using only the *scalar validation losses* already available in the loop, measuring each task's relative *progress* (current vs. initial validation loss) and giving more weight to the task that is improving more slowly. The overhead is negligible and no manual tuning is needed.

3.5 Pre-training setup (reference)

STELAR is pre-trained on a curated 500-hour subset of TUEG (0.3-75 Hz band-pass, 60 Hz notch, 200 Hz resample, 30 s windows, 19 channels, 50% masking) for 30 epochs with AdamW, peak learning rate 10^{-4} (warmup + cosine), batch size 64, on A100 GPUs. Table 2 summarizes the configuration.

Table 3: Improvement presets implemented on top of STELAR.

Preset	#1 attn-mask	#2 aug.	VICReg	Role
`v0_baseline`	—	—	—	faithful STELAR reference
`v1_attnmask`	✓	—	—	#1 isolated (diagnostic)
`v2_aug`	—	✓	—	#2 isolated (Option B)
`vA_combo`	✓	✓	—	#1 + #2 (Option A)

4 Our project: attempted improvements

Our extensions keep the STELAR backbone, objective and balancer *unchanged* and intervene only on two axes: *where* masking pressure is applied (#1) and *what* the encoder sees (#2). Crucially, neither adds a loss term, so MTPE-GB stays a two-task balancer; an optional anti-collapse term (VICReg) can be switched on as a third task if needed. All variants are exposed as named presets (Table 3).

4.1 Improvement #1: attention-guided masking

Idea. In STELAR, the 50% mask is random. We instead bias the mask toward *semantically salient* patches, estimated from the attention *received* in the last block of the teacher (which already runs a full forward to produce targets). The intuition is that masking informative regions makes the prediction task harder and richer than masking arbitrary ones.

Why teacher attention and not signal variance. Variance conflates semantics with subject idiosyncrasy; masking by variance would push the encoder toward subject-specific signatures and hurt generalization. Attention received is a cleaner semantic-saliency proxy.

Mechanics. We combine the two Criss-Cross attention maps (spatial and temporal), each min-max normalized, into an importance map over patches (gradient detached, it is a sampling prior, not a trained signal). We then sample the mask via *Gumbel-top-k* with temperature τ:

$$\text{perturbed} = \log(\text{importance}) + \tau \cdot \text{Gumbel}, \qquad \text{mask} = \text{top-}k(\text{perturbed}). \tag{2}$$

High τ makes noise dominate ($\approx$ uniform random masking); low τ concentrates on salient patches. A *cosine curriculum* decays τ from τ_{start}=1.0 to a non-zero floor $\tau_{\min}$=0.1 over $\sim$15 epochs, so training starts easy and gradually focuses. Importance is extracted by non-invasive forward-hooks on the teacher's last block, and the method *falls back to uniform masking* on any failure (missing modules, shape surprise, NaNs), so it degrades gracefully.

4.2 Improvement #2: asymmetric augmented views

Idea (data2vec-like asymmetry). The teacher encodes a *weak / clean* view (stable, clean latent targets), while the student encodes a *strong-augmented + masked* view and must predict the clean targets. Invariance to acquisition nuisances then emerges *inside* $\mathcal{L}_{\text{lat}}$, with **no extra loss term** (MTPE-GB stays two-task).

Augmentations (raw signal space). Augmentations are applied to the model input *before* the patch embedder, so that *both* the convolution (time) and FFT (frequency) branches see them. The student menu uses moderate amplitudes appropriate for low-SNR EEG: per-channel amplitude scaling $\sim \mathcal{U}(0.8, 1.2)$, additive noise, temporal jitter, crop-resize, and channel-dropout ($\leq$20%, zeroed in the *raw* signal). We deliberately *exclude* label-/montage-destroying transforms (time-reversal, sign-flip, channel permutation) and keep *band-dropout off* by default (it harms band-specific tasks and the FFT branch). Reconstruction and latent targets are always computed on the *clean* signal.

Table 4: Preliminary validation metrics (30 epochs, CPU, 10% data slice, single seed). Differences across presets are within noise; this run validates the pipeline, not the relative merit of the improvements.

Metric (validation)	`v0_baseline`	`v1_attnmask`	`v2_aug`
Best total loss (epoch)	0.7428 (ep5)	0.7413 (ep4)	**0.7410** (ep4)
Final total loss (ep29)	0.8116	0.8120	0.8114
Latent loss (first → last)	0.998 → 0.042	0.995 → **0.038**	0.998 → 0.040
Recon loss (first → last)	1.293 → 0.996	1.292 → 0.996	1.293 → 0.996
Student emb. std (first → last)	0.894 → 0.243	0.893 → 0.236	0.883 → 0.228
Teacher emb. std (first → last)	0.869 → 0.198	0.869 → 0.191	0.869 → 0.194
Collapse	none	none	none

4.3 Optional VICReg anti-collapse term

Stacking strong augmentation (#2) and hard masking (#1) raises the risk of *representation collapse*. Following standard practice, we always *log* the per-dimension embedding standard deviation of both teacher and student; if it collapses, an optional VICReg *variance hinge* on the *student* embeddings (which carry gradient) can be enabled as MTPE-GB's third task. Covariance/invariance VICReg terms are intentionally omitted: invariance is already provided by $\mathcal{L}_{\text{lat}}$, and the variance hinge alone is the cheap anti-collapse guard.

4.4 Decision protocol

Additivity of #1 and #2 is a hypothesis, not a guarantee, so we follow an empirics-first protocol: run #1-alone and #2-alone with identical seeds/splits; if #1-alone barely helps, do *not* combine and ship Option B (#2); if both clearly help, combine into Option A and tune *sequentially* (fix #2 at moderate defaults, search #1's τ schedule, then lightly retouch #2), never jointly.

5 Preliminary empirical study

Goal. Before any large-scale comparison, we validated that the three model scripts (`v0_baseline`, `v1_attnmask`, `v2_aug`) train end-to-end and behave sanely. This is a *smoke-scale* study, not a performance benchmark.

Setup. We trained on a single LMDB (a 10% validation slice of the pre-processed TUEG pipeline: 4,674 windows of shape $[19, 30, 200]$, i.e. only ~ 2 GB of data), split 90/10 into train/validation *on the fly*. All three presets, the baseline *and* both proposed improvements (`v1_attnmask`, `v2_aug`), were trained on this same slice, with the `tiny` configuration (encoder hidden 64, depth 2; $\sim$304K total parameters), batch size 64, 30 epochs, seed 7, learning rate 10^{-4}. Only the preset changed between runs. This very small scale was *forced by GPU limitations*, not chosen: runs were executed on whatever compute was available at the time (CPU or a small GPU, depending on the run), which is why we could not pre-train the improvements on anything larger than this $\sim$2 GB slice.

Results. Table 4 reports validation metrics. The three variants behave almost identically. In all of them the *latent* alignment loss falls from ≈ 1.0 to ≈ 0.04 (the student matches the EMA teacher well), the *reconstruction* loss plateaus near 0.996 (the tiny MLP reconstructor saturates at this scale), and the embedding standard deviation contracts from $\approx$0.87 to $\approx$0.20 but *stays clearly positive*, i.e. **no collapse**. The attention curriculum of #1 anneals as designed ($\tau : 1.0 \to 0.1$). The best total validation loss differs by only $\sim$0.2% across presets, which is within noise.

Reading the loss dynamics. The total loss *rises* after epoch 4-5 and then plateaus; this is *not* a regression. As the latent loss collapses toward zero, MTPE-GB shifts weight toward reconstruction ($w_{\text{rec}} : 0.50 \to 0.81$); the weighted sum therefore increases even though both individual losses are flat or still decreasing. This is exactly the adaptive behavior MTPE-GB is designed to produce.

Interpretation. All three pipelines train cleanly, converge, and show no collapse, so the engineering is validated. However, #1 and #2 show *no measurable advantage* over the baseline here. This is expected: a `tiny` model on a 10% data slice for 30 CPU epochs is far too small a regime for these improvements to express themselves, and the dominant dynamics (latent loss $\to 0$, reconstruction plateau, MTPE-GB reweighting) swamp any subtle effect. Consistent with the decision protocol, the empirics at this scale say "no signal yet" rather than picking a winner.

Threats to validity. Single seed, limited compute (CPU / a small GPU), a small $\sim$2 GB in-distribution slice (measuring fit, not transfer), and a tiny model. None of these allow conclusions about the improvements' true effect; they only establish correctness and stability.

6 Project organization: two complementary axes

Beyond the objective-level improvements of Section 4, our project work was organized around two further axes. Foundation models aim to learn *general* EEG representations that can be reused across many downstream tasks: instead of training a model from scratch for every dataset, the encoder is pre-trained on large, diverse data and then adapted by fine-tuning. This is especially relevant for EEG, where labeled data are limited, noisy and highly variable across subjects, devices, protocols and montages. Because EEG is high-dimensional, non-stationary and strongly subject-dependent, adapting such a model is not only a matter of fine-tuning the final classifier, but also of reconsidering the backbone, temporal patches, channels, positional information and inter-subject variability. Accordingly, the first axis (Section 7) studies modifications to the STELAR *backbone*, and the second (Section 8) studies *downstream* fine-tuning and evaluation.

7 Axis 1: architectural exploration of STELAR

The first axis reconsiders *how a foundation model should represent an EEG signal*, rather than merely tuning hyperparameters. EEG has a particular structure: a temporal axis, cut into patches, and a channel axis corresponding to electrodes on the scalp. Channels are therefore not independent dimensions (they also carry *spatial* information), which is why the choices around temporal patches, channel embedding, positional encoding and attention matter.

We considered two main directions. The first concerns *channel information*: whether the model should learn a representation specific to each channel, or instead be given more flexibility to adapt to different EEG montages. The second concerns *positional encoding*: the model must retain a notion of position in time, but potentially also a notion of spatial structure between electrodes. These directions both motivate the architectural choices already present in STELAR (learnable channel embeddings, RoPE in time) and frame possible extensions toward montage-agnostic and spatially-aware encoders. A recent example of this direction is REVE [3], which introduces a 4D positional encoding that lets a single pre-trained model process EEG of arbitrary length and electrode arrangement, pre-training on over 60,000 hours of data from 25,000 subjects to generalize across heterogeneous montages.

8 Axis 2: downstream evaluation via linear probing and fine-tuning

The decisive metric for an EEG encoder is not pre-training loss but *downstream transfer*. We assess it by *linear probing* (freezing the encoder and training only a classification head, which

Table 5: Preliminary downstream results on FACED and TUEV (best available metrics; "—" = not recorded). Accuracy, balanced accuracy and macro F1 in %; Cohen's κ as a fraction. Best per dataset in **bold**.

Dataset	Configuration	Acc	Bal-Acc	Macro-F1	κ
FACED	Pre-trained, *probe* head (best)	**47.4**	**47.5**	**47.4**	**0.407**
	Frozen encoder	43.3	—	—	0.362
	Top-2 block fine-tune	43.4	—	—	0.363
	average head + top-2 FT	18.3	—	—	0.072
TUEV	Stronger encoder LR (best)	**70.0**	**42.9**	**41.5**	**0.419**
	Baseline fine-tune	67.8	—	—	0.388
	Frozen encoder	66.5	—	—	0.369
	Top-2 block fine-tune	67.2	—	—	0.375

isolates the intrinsic quality of the representation) and, more generally, by *fine-tuning* (letting the encoder adapt). The targeted tasks were CSBrain, FACED, KaggleERN, PhysioMI, Sleep-EDFx, TUAB and TUEV; the finalized results concern **FACED**, **TUEV** and **BCIC-2A**, plus a dedicated fine-tuning study on **TUAB**, **PhysioMI** and **KaggleERN** (Section 8.3).

The general principle is the standard transfer-learning setup: start from a pre-trained encoder (the baseline checkpoint) and adapt it to a target task by adding a task-specific head: the encoder provides a general EEG representation, and the head learns the mapping to the dataset labels.

Adaptation strategies. We tested several. With a *frozen* encoder, only the head is trained; good performance then means the pre-trained representation already contains task-useful information. Alternatively, the encoder is updated during training, either *fully* (better adaptation, but a higher overfitting risk on small EEG datasets) or *partially* (e.g. unfreezing only the last few blocks) as a compromise that preserves pre-trained knowledge while allowing some task adaptation.

Protocol. We used the best available baseline checkpoint and ran the downstream scripts across configurations: baseline fine-tuning, frozen encoder, partial top-k-block fine-tuning, and in some cases a variation of the encoder learning rate. Although this partly amounts to running scripts, it defines the experimental frame for comparing variants: without a baseline it is hard to know whether observed changes come from architectural modifications or from the model's normal fine-tuning behavior. We report accuracy, balanced accuracy, macro F1 and Cohen's κ; reporting balanced accuracy and κ alongside accuracy matters because, under class imbalance, a model can show decent accuracy while recognizing some classes poorly.

8.1 Results on FACED and TUEV

On **FACED** (Table 5, top), the choice of the downstream *head* had a large impact. The best configuration (a pre-trained STELAR with a *probe*-type head) reaches ∼47% accuracy and $\kappa = 0.407$. A frozen encoder ($\kappa = 0.362$) and top-2-block fine-tuning ($\kappa = 0.363$) are very close, so partial fine-tuning brought no clear gain over freezing here. Strikingly, an *average* head with top-2 fine-tuning collapses to ∼18% accuracy ($\kappa = 0.072$): the head is not a detail and can matter almost as much as the fine-tuning strategy.

On **TUEV** (Table 5, bottom), the best configuration uses a *stronger encoder learning rate* (∼70% accuracy, $\kappa = 0.419$), above the baseline ($\kappa = 0.388$), the frozen encoder ($\kappa = 0.369$) and top-2 fine-tuning ($\kappa = 0.375$). Letting the encoder adapt more strongly is thus more useful here than freezing or fine-tuning only two blocks. However, the gap between accuracy (∼70%) and balanced accuracy (∼43%) is large, indicating class imbalance or some much harder classes: accuracy alone is not enough to conclude the model is good on all classes.

Table 6: Preliminary BCIC-2A results (leave-one-subject-out). DANN = our domain-adversarial extension (Section 9). Accuracy in %, κ as a fraction. Best in **bold**.

Configuration (LOSO)	Acc	κ
STELAR patch-adapt (pre-trained)	41.0	0.213
DANN, random init, frozen encoder	**54.2**	**0.390**
DANN, pre-trained frozen encoder (STELAR)	52.2	0.362
DANN, pre-trained frozen encoder (EEGPT)	51.4	0.352

8.2 Results on BCIC-2A

On **BCIC-2A** (Table 6), inter-subject generalization is very demanding. The pre-trained STELAR patch-adapt configuration, in leave-one-subject-out (LOSO), gives only $\sim$41% accuracy ($\kappa = 0.213$): transfer is not direct in this setting. Domain-adversarial variants (DANN, see Section 9) are higher on average, up to $\sim$54% accuracy ($\kappa = 0.390$) with random initialization and a frozen encoder, and $\sim$52%/$\sim$51% for pre-trained frozen STELAR/EEGPT encoders. This comparison warrants caution: the random-init version being better does not necessarily mean pre-training is useless, it may reflect the configuration, the head, the frozen encoder or the split protocol. More honestly, BCIC-2A remains a hard inter-subject task and the DANN results deserve a cleaner study.

8.3 A dedicated fine-tuning pipeline and further results

We also ran a more systematic fine-tuning study from a lab-pre-trained STELAR checkpoint, with subject-wise cross-validated scores on four further datasets.

Checkpoint. The checkpoint is a `little` STELAR encoder (embedding dimension 128, 2 heads, depth 8), pre-trained for 50 epochs on the clinical TUEG corpus ($\sim$500 h) at a 70% masking ratio, with a final validation loss of 0.103. Crucially, **TUEV and TUAB were deliberately excluded** from pre-training, so that downstream evaluation on them is uncontaminated.

Pipeline. Each window is (1) re-channeled by a learned channel convolution mapping the dataset's N electrodes (e.g. 32 for FACED, 64 for PhysioMI) onto STELAR's 19, (2) normalized (divided by 100), (3) split into temporal patches (100 points for FACED, 50 for PhysioMI), encoded by the dual-branch Criss-Cross encoder with temporal RoPE, and mean-pooled into a 128-dimensional window embedding, then (4) classified by an EEGPTProbeHead (a norm-constrained two-layer MLP). The channel module and head are trained from scratch; the encoder is only lightly adapted, with separate learning rates (2.5×10^{-3} for the head, 1×10^{-4} for the encoder), OneCycleLR, and gradient clipping at 1.0.

Protocol and results. Evaluation is strictly *subject-wise*: data are partitioned by subject so that no window from a test subject is ever seen in training. This is indispensable in EEG: otherwise the model would recognize the subject rather than the task, inflating the scores. We use 4 folds for FACED, TUAB and TUEV and 5 folds for PhysioMI and KaggleERN, and report fold-averaged metrics; runs used SLURM on Telecom Paris P100 GPUs (16 GB), 2 to 6 hours each. Table 7 reports the scores. **TUAB** is the strongest at 79.6% balanced accuracy, a clinically central normal-versus-abnormal task the encoder never saw during pre-training; **FACED** reaches 47.4% accuracy, consistent with Table 5; **PhysioMI** (64.2%) and **KaggleERN** (55.6%) are harder, as expected for fine-grained motor-imagery and error-related paradigms.

Table 7: Additional fine-tuning results from the lab-pre-trained STELAR `little` checkpoint, under subject-wise cross-validation (fold-averaged).

Dataset	Main metric	Score (%)
FACED	Accuracy (4-fold)	47.4
TUAB	Balanced accuracy (4-fold)	**79.6**
PhysioMI	Balanced accuracy (5-fold)	64.2
KaggleERN	Balanced accuracy (5-fold)	55.6

9 Subject-invariant representations: a DANN extension

An early intuition was that some of STELAR's limits may stem from *subject overfitting*, natural in EEG, where signals are strongly subject-dependent: a model may latch onto subject identity instead of task-relevant information. We therefore considered a DANN-inspired extension (Domain-Adversarial Neural Network): keep the main head predicting the EEG class, add a second head that predicts the *subject*, and connect it through a *gradient-reversal layer* so the encoder is pushed to produce representations that are useful for the task but uninformative about subject identity.

The goal was twofold: (i) *regularization*, to reduce subject overfitting; and (ii) *diagnosis*, since a discriminator that easily recovers the subject reveals subject-specific information in the encoder, whereas reducing it without hurting the main task suggests better generalization. Early BCIC-2A results are encouraging, DANN configurations beat the plain patch-adapt setup, but not yet conclusive: with a *frozen* encoder the adversarial loss cannot change the representations, which strongly limits interpretation. Properly validating the idea requires training the encoder with the adversarial loss *active*, sweeping the trade-off weight λ, monitoring the subject-discriminator loss, and checking that main-task performance does not collapse, across several seeds and subject-wise protocols. For now, DANN is a promising direction and a useful lens on overfitting and subject dependence.

10 General observations and limitations

Overall, downstream transfer depends heavily on the dataset and the adaptation strategy: on FACED the *head* dominates; on TUEV *stronger* encoder fine-tuning helps; on BCIC-2A inter-subject generalization stays hard and DANN is promising but hard to read. A second limitation is computational cost: testing all variants properly (frozen, full, top-k, several heads, learning rates, datasets and seeds) demands substantial GPU time, so the present comparisons are partial. Finally, the results confirm that adapting a foundation model to EEG is not merely reusing a checkpoint: the classification head, the split protocol, the metrics and the fine-tuning strategy must all be chosen carefully. STELAR shows real potential for downstream EEG, but its robustness depends strongly on how it is adapted to each dataset.

Our results in the broader debate on EEG foundation models. Our findings echo a recent, more systematic critique of the field. Benchmarking state-of-the-art EEG foundation models against compact neural networks and classical non-neural decoders, Yang et al. [4] report that (i) *linear probing remains consistently weak and highly task-dependent*; (ii) foundation models *do not universally outperform* simple baselines, often falling behind in data-scarce or subject-specific settings; (iii) there is *no clear scaling law*, so larger models are not reliably better; and (iv) *simple pre-training can be effective*, since a plain masked auto-encoder on raw EEG can already be top-performing. Our own small-scale experiments are consistent with this picture: linear-probe transfer was modest and brittle, the classification head and adaptation protocol mattered as much as the encoder, and inter-subject generalization (BCIC-2A) stayed hard. This frames our improvements not as a guaranteed win but as one probe into *which* ingredients of a

dual-space objective actually help, a question that, as Yang et al. [4] argue, can only be answered with large standardized EEG corpora and statistically rigorous, multi-protocol evaluation.

11 Conclusion and future work

We reviewed the EEG context and the self-supervised state of the art, explained the STELAR dual-space foundation model and its validation-driven MTPE-GB balancer, and implemented two targeted improvements, attention-guided masking and asymmetric augmented views, plus an optional anti-collapse term, all without modifying the backbone or adding loss terms. A preliminary small-scale study confirmed that every variant trains end-to-end without collapse, while, as expected at smoke scale, showing no measurable separation between them. We further explored architectural directions for the backbone (channel handling and positional encoding) and ran a preliminary downstream evaluation on FACED, TUEV and BCIC-2A: transfer depends strongly on the fine-tuning strategy and the classification head, and inter-subject generalization (BCIC-2A) remains the hardest case, where a domain-adversarial (DANN) extension looks promising.

The clear next steps are: (i) pre-train each variant at an adequate scale (larger model, full corpus, GPU); (ii) extend the downstream evaluation of Section 8 across all seven tasks and several seeds; (iii) train the DANN extension with the adversarial loss *active* (not a frozen encoder), sweeping λ, to test whether reducing subject information improves cross-subject transfer; and (iv) if and only if #1 and #2 each help in isolation, combine them sequentially into Option A while monitoring the embedding-std curves for collapse. Only then can the improvements be accepted or rejected on evidence.

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